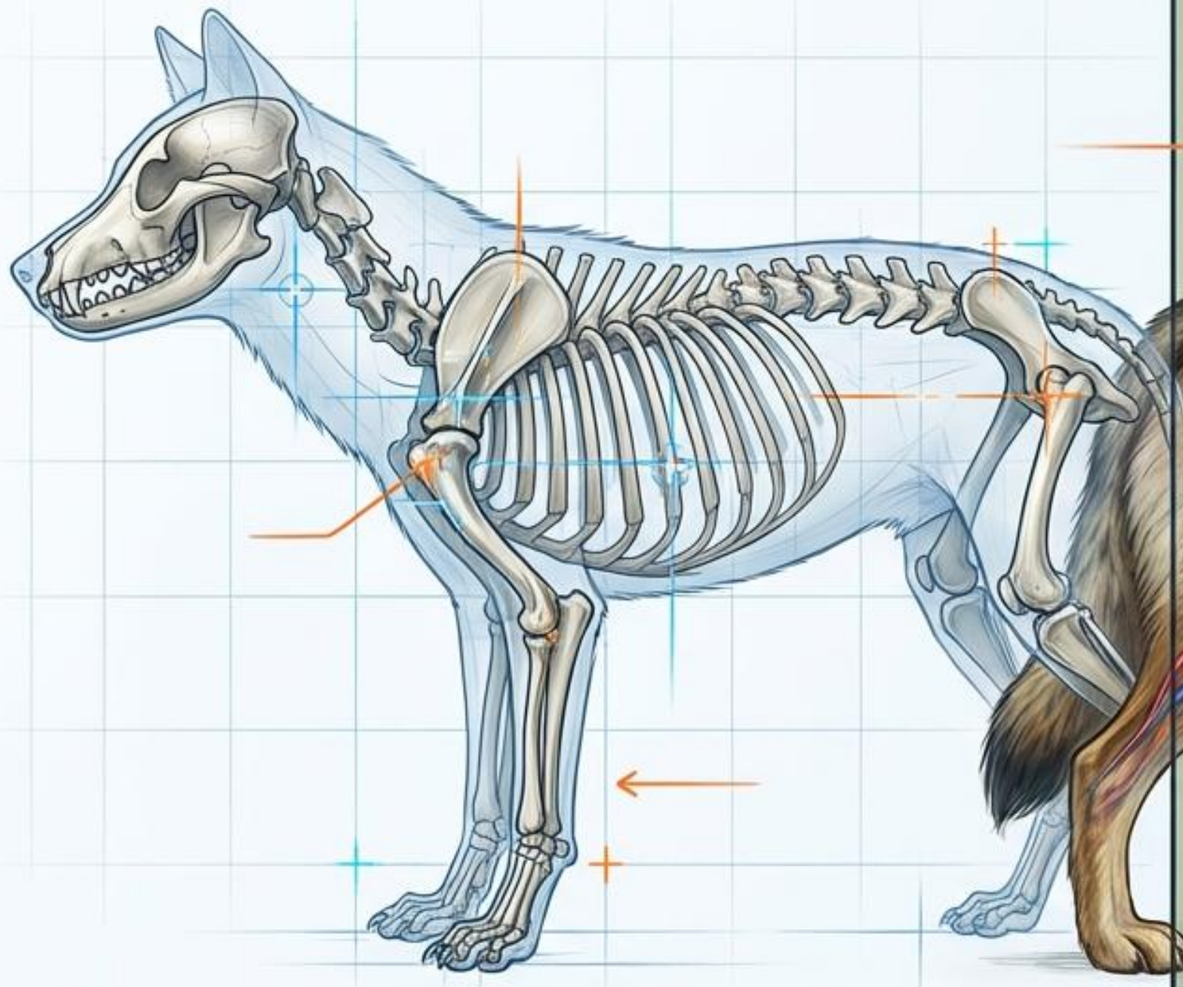
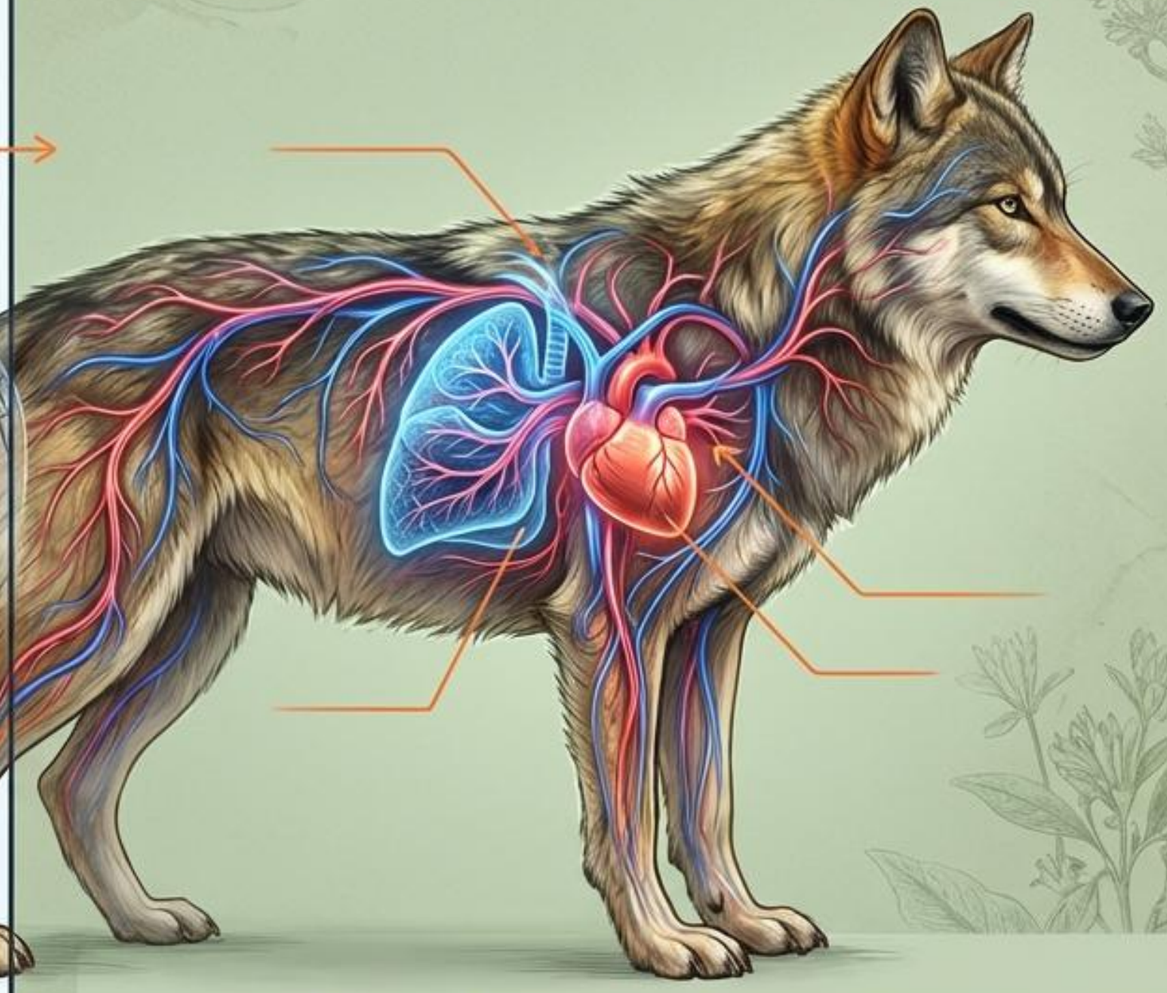


Anatomy [The Hardware]:
The physical structures that make up an organism.

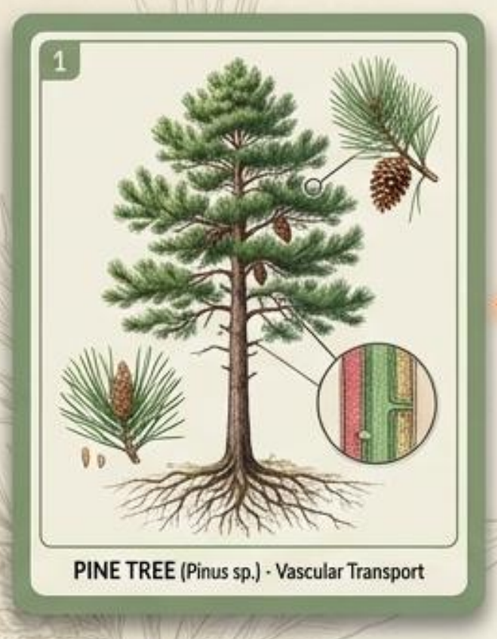


Physiology [The Software]:
The dynamic processes and functions those structures perform to keep the organism alive.

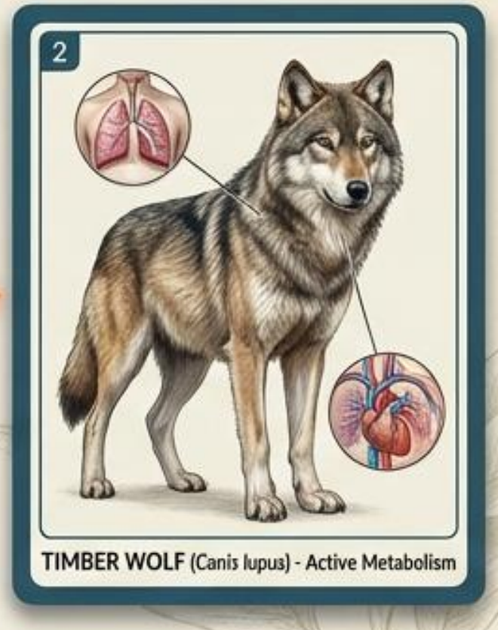
NotebookLM

Universal Problems, Diverse Solutions

Every living organism faces the exact same fundamental challenges.
Anatomy and physiology are the evolutionary solutions.

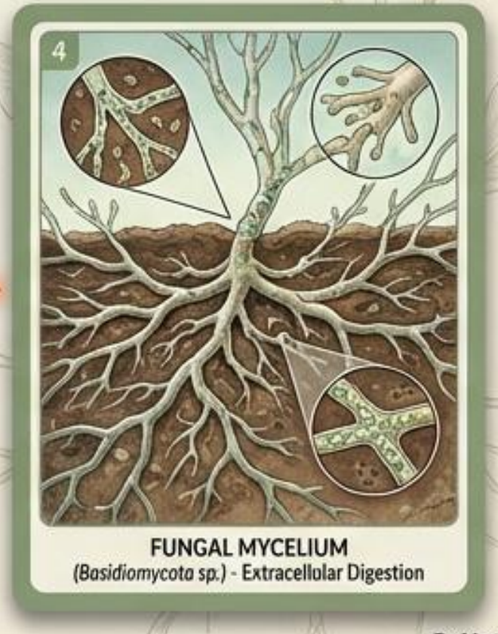


PINE TREE (Pinus sp.) - Vascular Transport



TIMBER WOLF (Canis lupus) - Active Metabolism

KELP FROND
(Macrocystis pyrifera) - Aquatic Adaptations



FUNGAL MYCELIUM
(Basidiomycota sp.) - Extracellular Digestion

A pine tree and a timber wolf must both transport water and nutrients, but their internal machinery evolved on entirely different paths.

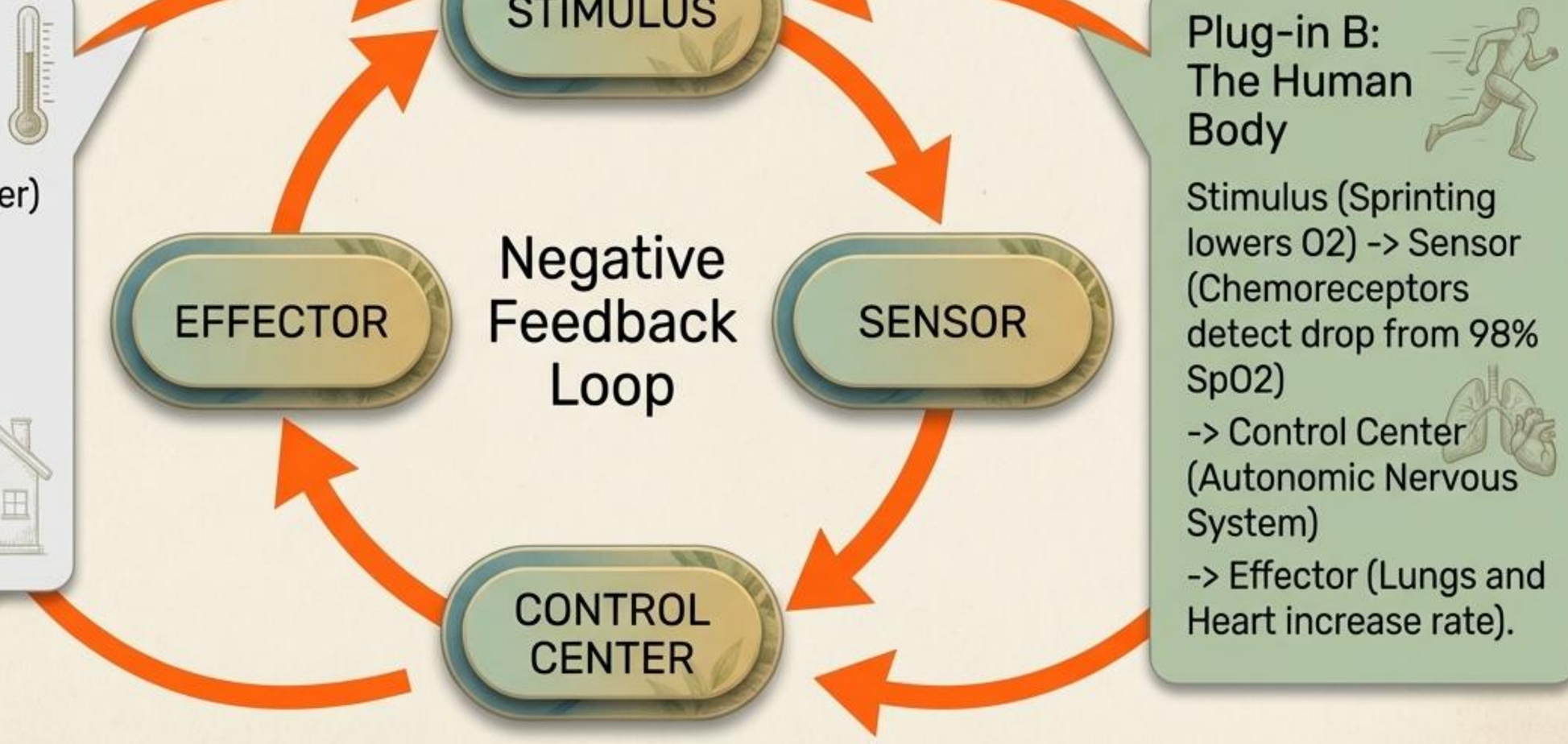
NotebookLM

The Homeostasis Engine (Negative Feedback)

Homeostasis is the maintenance of a stable internal environment despite external chaos.

Plug-in A: The Thermostat

Stimulus (Cold Winter) -> Sensor (Thermometer reading 21°C) -> Control Center (Thermostat) -> Effector (Heater turns on).



Plug-in B: The Human Body

Stimulus (Sprinting lowers O2) -> Sensor (Chemoreceptors detect drop from 98% SpO2)

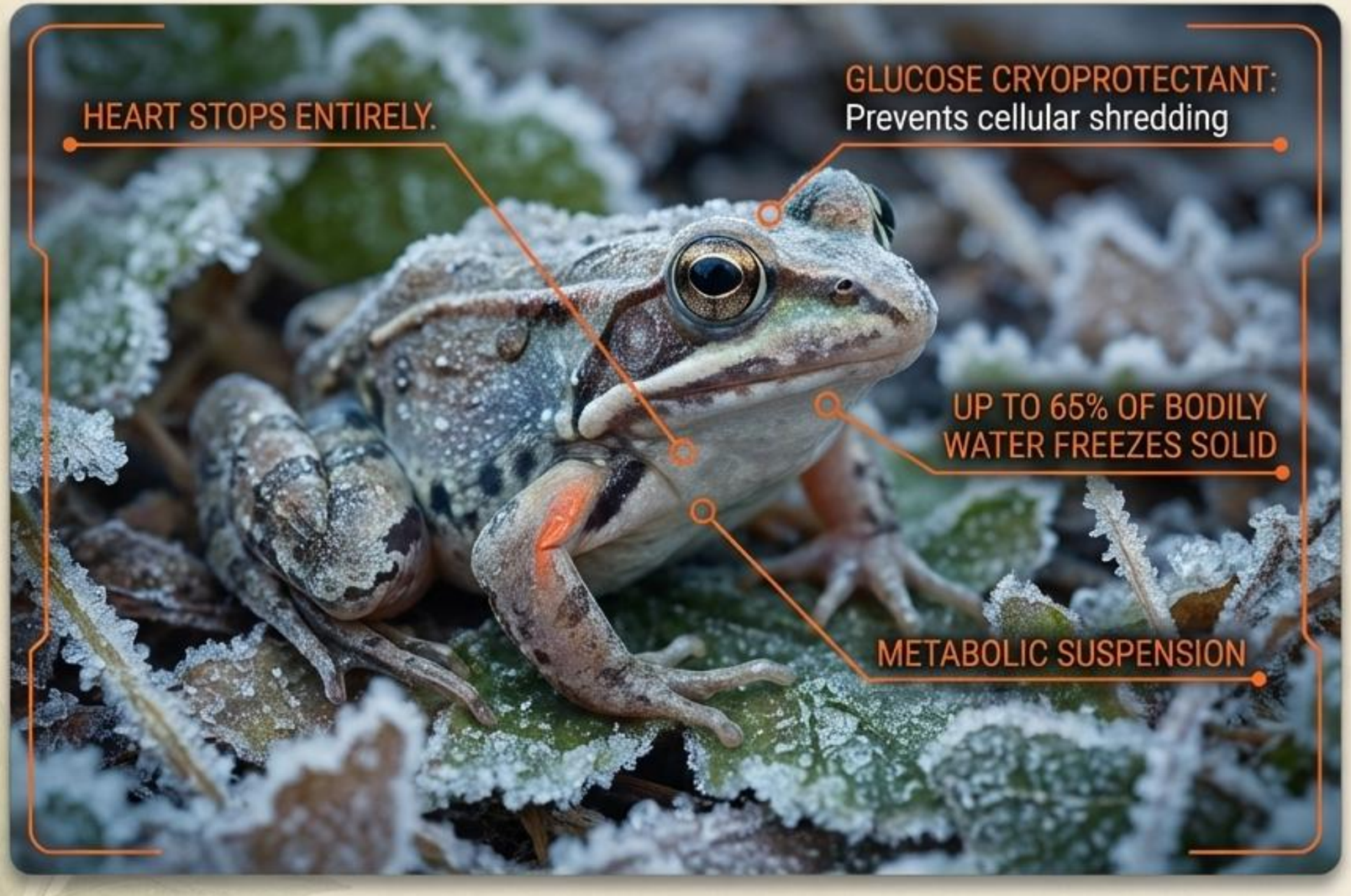
-> Control Center (Autonomic Nervous System)

-> Effector (Lungs and Heart increase rate).

NotebookLM

Field Note: Extreme Thermoregulation

Maintaining homeostasis during a Saskatchewan winter requires radical physiological adaptations.



The Endotherm Approach (Arctic Fox):

Generates heat internally via increased metabolism and traps it with a thick winter coat.

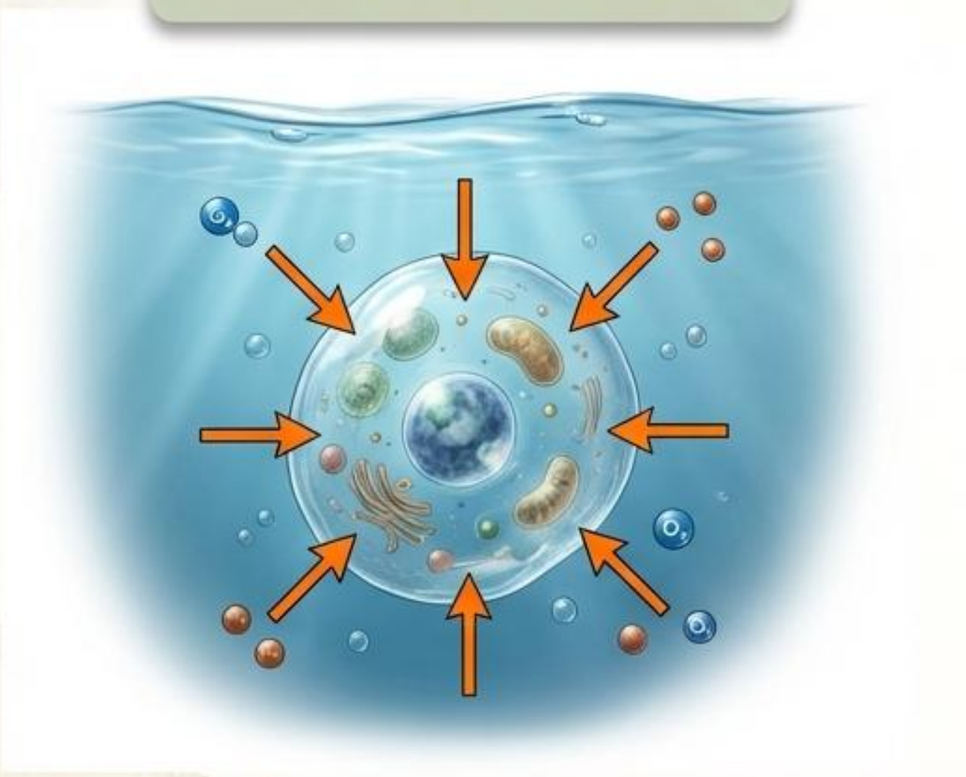
The Ectotherm Anomaly (Wood Frog):

- Heart stops entirely.
- Up to 65% of bodily water freezes solid.
- **The Antifreeze**: Floods tissues with glucose (sugar) acting as a biological cryoprotectant to prevent sharp ice crystals from shredding cellular structures.

NotebookLM

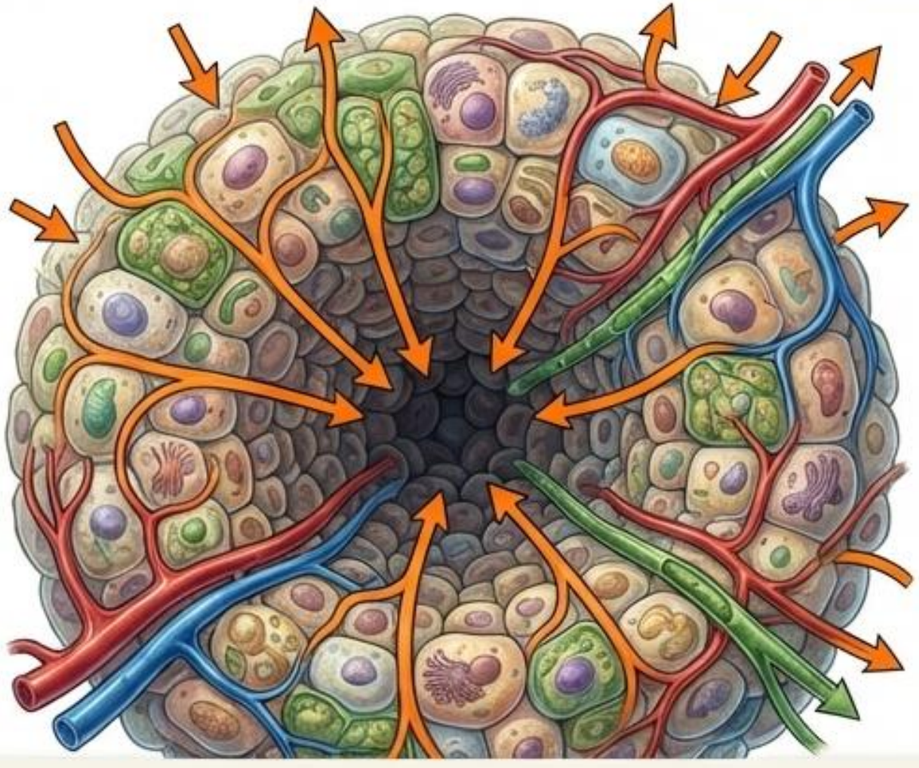
The Size Problem demands specialization

DIRECT DIFFUSION



Direct Diffusion: Single-celled protists rely on immediate contact with their environment for gas exchange and waste removal.

INTERNAL HIGHWAYS



Internal Highways: Multicellularity creates a profound problem: internal cells are isolated. Survival requires the evolution of specialized tissues and organs for transport, digestion, and support.

NotebookLM

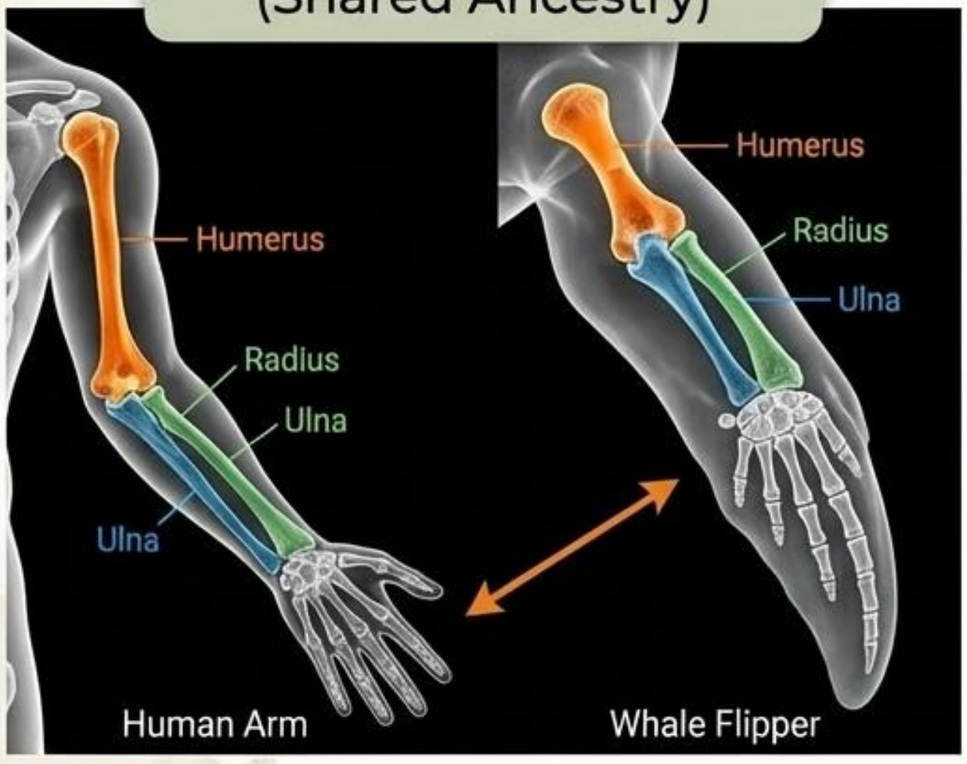
Cross-Kingdom Anatomy Matrix

	Transport	Digestion	Gas Exchange
Protists (e.g., Kelp)	None (water environment).	Intracellular.	Direct diffusion.
Fungi	Mycelial networks.	Extracellular (secreting enzymes via hyphae to absorb broken-down nutrients).	Direct diffusion.
Plantae	Vascular plumbing (Xylem for water up, Phloem for sugars down).	Autotrophic (Photosynthesis).	Microscopic stomata in leaves.
Animalia	Circulatory system (heart/vessels).	Complex internal digestive tracts.	Specialized respiratory organs (lungs, gills).

NotebookLM

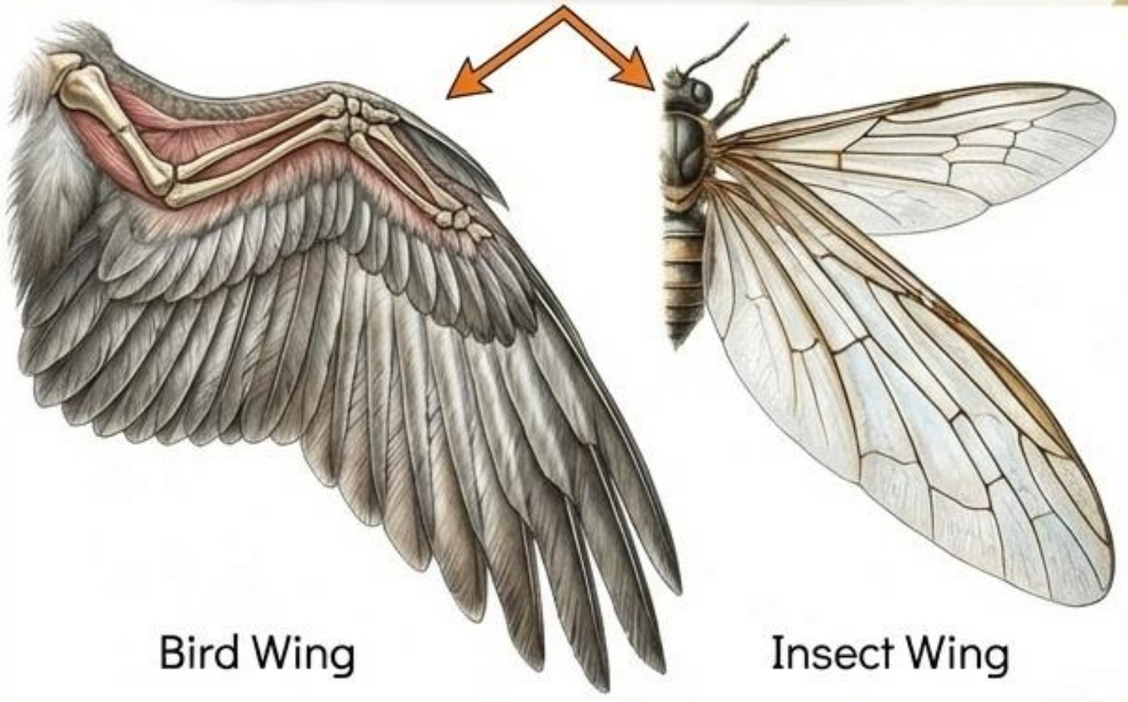
Tracing Evolutionary Origins

Homologous Structures (Shared Ancestry)



Homologous Structures (Shared Ancestry): Similar underlying anatomical framework adapted for entirely different functions (e.g., grasping vs. swimming).

Analogous Structures (Shared Environment)



Analogous Structures (Shared Environment): Different underlying anatomies that evolved independently to solve the exact same environmental pressure (e.g., achieving flight).

NotebookLM

The Lab Roster: Model Organisms

Biologists study non-human species to unlock the biological phenomena of all life, chosen for short generation times and genetic similarities (mice share 85% of human protein-coding genes).

E. coli
(Bacterium)

Thale cress
(Plant)

Fruit fly
(Insect)

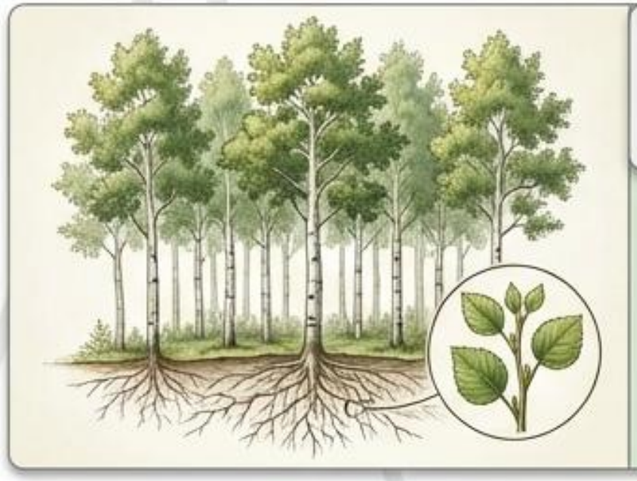
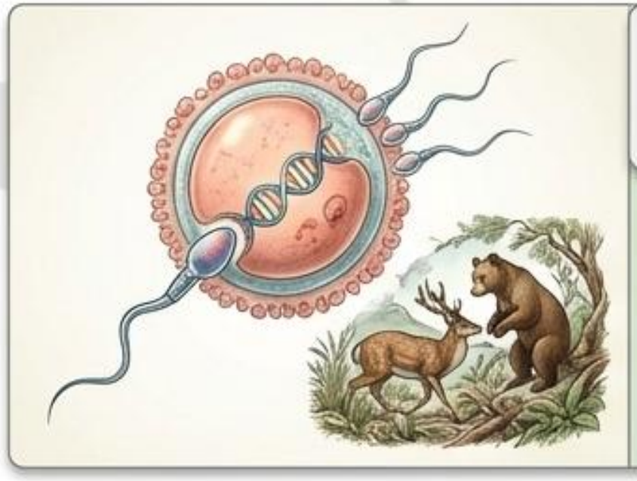
House mouse
(Mammal)

The Ethical Imperative (The 3Rs):

- **Replacement:** Can we use a computer simulation or cell culture instead?
- **Reduction:** Can we use fewer animals for valid results?
- **Refinement:** Can we modify procedures to eliminate distress?

NotebookLM

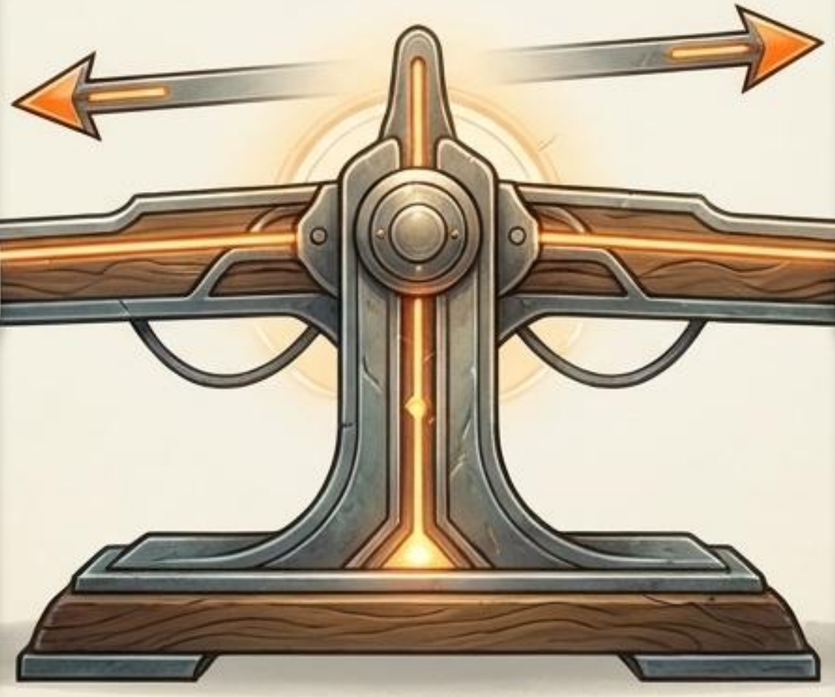
The Reproductive Trade-off

	Evolutionary Benefit	Evolutionary Cost
	**Asexual Reproduction** (e.g., Trembling Aspen clonal colonies) Mechanism: Single parent produces genetically identical clones.	
	✓ Extremely fast. ✓ Zero energy spent finding a mate. Ideal for stable environments.	⚠ Zero genetic diversity. ⚠ A single disease or climate shift can wipe out the entire clone population.
	**Sexual Reproduction** Mechanism: Two parents combine genetic material (sperm/egg).	
	✓ Massive genetic diversity. ✓ Provides the variation necessary to adapt to a changing environment.	⚠ Highly energy-intensive. ⚠ Finding mates is dangerous, and only half of an individual's genes are passed on.

NotebookLM

The Survival Spectrum:
r/K Selection Theory

"Even within sexual reproduction, species diverge radically in how they allocate energy to ensure their lineage survives."



r

r-Selected Species (Quantity): Produce massive numbers of offspring with zero parental care. Most die, but sheer numbers guarantee survival.

Example: Northern Pike (100,000 eggs per female).

K

K-Selected Species (Quality): Produce very few offspring but invest massive amounts of energy and time into parental care, ensuring a high survival rate.

Example: Black Bear (2 cubs raised for up to two years).

NotebookLM

The Defense Arsenal

To survive the endless arms race between predator and prey, organisms deploy three types of defenses:

Structural Defenses (Physical Barriers)

The quills of a porcupine, the thick bark of an oak tree, or the waxy cuticle of a leaf blocking bacteria.

Behavioural Defenses (Evasive Actions)

The Killdeer bird faking a broken wing to draw predators away from its nest, or nocturnal desert feeding.

Physiological Defenses (Internal Chemistry)

The targeted antibodies of a complex immune system, snake venom, or plants synthesizing toxic alkaloids.

NotebookLM

Synthesis: The Interconnected Prairie

All diverse evolutionary solutions interact in one shared environment. Test your recall:

2. Which organism uses complex compartmentalized lungs to support a high metabolism? (Animalia/Endotherm)

1. Where is extracellular digestion happening? (Fungi recycling nutrients)

3. Where are hollow vascular tubes transporting oxygen down to submerged roots? (Plantae)

Water molds (Fungi)

NotebookLM